

# Upgrade Notes — v17.58: Sequence State Machine & BOS-Anchored Ranges

**⚠ BREAKING CHANGE (indicator side).** The v17.58 Pine Script reworks the alert contract around a 5-state institutional sequence and BOS-anchored swing ranges. **The Railway backend remains fully backward-compatible** — older payloads (v17.57 / v17.56.9 / v17.56.8 / v17.56.7 / legacy) continue to work unchanged, with the new fields defaulting safely. The “breaking” part is the indicator’s output schema; the backend gracefully accepts both.

## Overview

v17.58 introduces two related institutional concepts to the OIE webhook payload:

1. **Sequence State Machine** — the indicator now drives a deterministic 5-state sequence describing how a setup matures:

| State | Step         | Meaning                                   |
|-------|--------------|-------------------------------------------|
| 0     | idle         | no active sequence                        |
| 1     | LOCATION     | price is in the HTF Premium/Discount zone |
| 2     | LIQUIDITY    | liquidity has been swept                  |
| 3     | DISPLACEMENT | LTF shift / displacement detected         |
| 4     | MITIGATION   | price returns into the mitigation zone    |
| 5     | EXECUTION    | entry trigger fired — sequence complete   |

1. **BOS-Anchored Ranges** — the active Break-of-Structure swing range ( `high` / `low` / `50% equilibrium` ) and its directional trend.

Unlike the v17.57 PDH/PDL levels (which were **in-memory only**), these 17 fields **ARE persisted** to the SQLite `signals` table, so there **is a schema migration** in this release (idempotent, runs automatically on deploy).

## New Fields (17)

### Sequence state machine (4)

| Field             | Type | Source                   | Default | Meaning                                        |
|-------------------|------|--------------------------|---------|------------------------------------------------|
| sequence_state    | int  | sequence_state<br>0-5    | 0       | Current state in the 5-state sequence          |
| sequence_step     | text | sequence_step            | NULL    | Human-readable current step label              |
| missing_step      | text | missing_step             | NULL    | Next/missing step to complete the sequence     |
| sequence_complete | bool | sequence_complete<br>0/1 | False   | Whether the full 5-state sequence is satisfied |

### BOS-anchored ranges (4)

| Field           | Type  | Source              | Default | Meaning                                               |
|-----------------|-------|---------------------|---------|-------------------------------------------------------|
| bos_range_high  | float | bos_range_high      | 0.0     | Active BOS swing high                                 |
| bos_range_low   | float | bos_range_low       | 0.0     | Active BOS swing low                                  |
| bos_equilibrium | float | bos_equilibrium     | 0.0     | 50% equilibrium of the BOS range                      |
| bos_trend       | int   | bos_trend<br>1/-1/0 | 0       | BOS direction:<br>1=bullish,<br>-1=bearish,<br>0=none |

## State completion flags (5)

| Field               | Type | Source                  | Default |
|---------------------|------|-------------------------|---------|
| state1_location     | bool | state1_location 0/1     | False   |
| state2_liquidity    | bool | state2_liquidity 0/1    | False   |
| state3_displacement | bool | state3_displacement 0/1 | False   |
| state4_mitigation   | bool | state4_mitigation 0/1   | False   |
| state5_execution    | bool | state5_execution 0/1    | False   |

## Liquidity / shift detection (4)

| Field                 | Type | Source                    | Default |
|-----------------------|------|---------------------------|---------|
| liquidity_swept       | bool | liquidity_swept 0/1       | False   |
| ltf_shift_detected    | bool | ltf_shift_detected 0/1    | False   |
| displacement_detected | bool | displacement_detected 0/1 | False   |
| mitigation_zone       | bool | mitigation_zone 0/1       | False   |

Booleans arrive as integer `0 / 1` from Pine Script and are coerced to Python `bool` (stored as `0 / 1` in SQLite, which has no native `BOOLEAN`). `sequence_state` is validated to the domain `{0,1,2,3,4,5}` and `bos_trend` to `{1,-1,0}`; out-of-range or malformed values degrade gracefully to the defaults above rather than raising, via the existing safe helpers (`_to_float`, `_to_int`) plus the new `normalize_sequence_state` / `normalize_bos_trend` validators in `src/version.py`.

## Example v17.58 payload

```
{
  "version": "v17.58",
  "mode": "EXECUTION", "session": "NY", "symbol": "EURUSD",
  "direction": "LONG", "setup": "A+ SNIPER", "poi": 7,
  "entry": 1.0855, "sl": 1.0825, "tp": 1.0945, "valid": true,
  "sequence_state": 5, "sequence_step": "EXECUTION",
  "missing_step": "", "sequence_complete": 1,
  "bos_range_high": 1.0900, "bos_range_low": 1.0800,
  "bos_equilibrium": 1.0850, "bos_trend": 1,
  "state1_location": 1, "state2_liquidity": 1, "state3_displacement": 1,
  "state4_mitigation": 1, "state5_execution": 1,
  "liquidity_swept": 1, "ltf_shift_detected": 1,
  "displacement_detected": 1, "mitigation_zone": 1
}
```

## Database Migration

This release adds **17 columns to the** `signals` **table** plus two indexes:

- `idx_signals_sequence_state` (on `sequence_state` )
- `idx_signals_sequence_complete` (on `sequence_complete` )

The migration is implemented in

`src/database.py::_run_sequence_state_migration()` and is **idempotent** — it checks `PRAGMA table_info(signals)` before each `ALTER TABLE ... ADD COLUMN` , because SQLite (the Railway runtime DB) does **not** support `ADD COLUMN IF NOT EXISTS` . It is wired into `init_db()` and runs automatically on every deploy/startup. A canonical reference SQL script lives at `schemas/migrate_v17_58_sequence_fields.sql` .

**No data loss / no downtime:** existing rows get the column defaults (`0` / `0.0` / `NULL` ); re-running the migration is a no-op.

## API Changes

### New endpoint: `GET /api/v1/sequence-analytics`

Returns the distribution of persisted signals across the 5-state sequence, the completion count and completion rate. Supports an optional `?pair=` filter.

```

{
  "version": "v17.58",
  "pair": null,
  "total_signals": 42,
  "sequence_complete": 11,
  "completion_rate_pct": 26.19,
  "state_distribution": {
    "0": {"label": "idle", "count": 5},
    "1": {"label": "location", "count": 8},
    "2": {"label": "liquidity", "count": 7},
    "3": {"label": "displacement", "count": 6},
    "4": {"label": "mitigation", "count": 5},
    "5": {"label": "execution", "count": 11}
  }
}

```

## Extended endpoints

- GET /api/v1/signals — now accepts `?sequence_state=5` and `?sequence_complete=1` filters, and rows now include all 17 persisted fields.
- POST /api/v1/signal — response now echoes the sequence/BOS fields.
- GET /api/v1/latest — the in-memory snapshot now includes the v17.58 fields.
- GET /api/v1/health — advertises the new `sequence_state_machine` and `bos_anchored_ranges` feature flags.

## Backward Compatibility

| Payload version | Behaviour                                                            |
|-----------------|----------------------------------------------------------------------|
| v17.58          | All 17 fields parsed and persisted                                   |
| v17.57          | New fields default (0 / 0.0 / NULL / False); PDH/PDL still in-memory |
| v17.56.9        | New fields default; Guardian fields still parsed                     |
| v17.56.8        | New fields default; AMD/HUD fields still parsed                      |
| v17.56.7        | New fields default; dual-mode fields still parsed                    |
| legacy/v17.25   | New fields default                                                   |

The universal `decode_payload()` dispatcher routes `v17.58*` to the new `decode_v17_58_payload()` (which layers `_apply_sequence_bos_fields()` on top of `decode_v17_57_payload()`), and every older decoder also applies the new-field defaults so they are always present on the record.

## Files Modified

- `src/version.py` — `VERSION` → `v17.58`; added `bos_anchored_ranges` + `sequence_state_machine` feature flags; added `normalize_sequence_state` / `normalize_bos_trend` validators.
- `src/oie_processor.py` — parse the 17 fields in `normalize_oie_payload`; new `_apply_sequence_bos_fields()` helper; new `decode_v17_58_payload()`; default-application in all older decoders; new dispatcher branch; threaded fields through `oie_to_legacy_compact()`.
- `src/tracker/processor.py` — `expand_compact_entry()` extracts the 17 `_`-prefixed bridge fields into the persisted record.
- `src/database.py` — `SEQUENCE_STATE_MIGRATION_PATH`; `_run_sequence_state_migration()`; wired into `init_db()`; added the 17 columns to `insert_signal()` `known_columns`; `sequence_state` / `sequence_complete` filters in `get_signals()` / `count_signals()`.
- `src/webhook_server/routes.py` — `/sequence-analytics` endpoint; `/signals` filters; `POST /signal` echo; `/latest` cache fields; docstring.
- `app.py` — startup banner.
- `schemas/migrate_v17_58_sequence_fields.sql` — canonical migration reference.
- `test_v17_58.py` — new suite (155 checks).
- `test_v17_57.py` — relaxed 4 hardcoded `== "v17.57"` version asserts to `>=`.

## Test Results

```
test_v17_58.py ..... 155 passed, 0 failed
test_v17_57.py ..... 97 passed, 0 failed
test_v17_56_9.py ..... 66 passed, 0 failed
test_v17_56_8.py ..... 53 passed, 0 failed
test_oie_integration.py ... ALL INTEGRATION TESTS PASSED
```

## Deployment / Merge Order

This PR is **stacked** on the v17.57 PR. Railway auto-deploys from `main`, so merge the chain **in order** to keep history linear and migrations sequential:

```
#12 (v17.56.7) → #14 (v17.56.8) → #16 (v17.56.9) → #17 (v17.57) → #18 (v17.58)
```

On deploy, `init_db()` runs the idempotent migration automatically — no manual SQL step is required.